

#Atmospheric Dynamics



Chapter Five : Circulation and Vorticity



Objectives

- *To Distinguish Circulation and vorticity*
- *To be aware of the application of Circulation Theorem*
- *To know Various Forms of Vorticity*
- *To Derive vorticity equation*

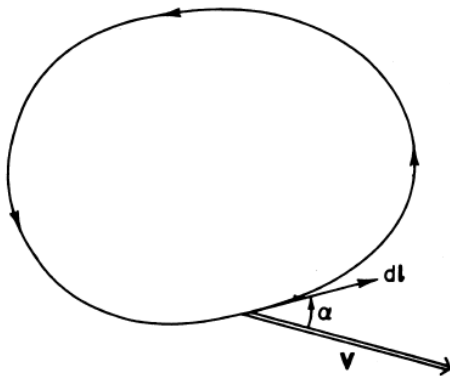
Circulation and Vorticity

- Circulation and vorticity are the two primary measures of rotation in a fluid.
- Circulation: is a scalar integral quantity and is a *macroscopic* measure of rotation for a finite area of the fluid.
- Vorticity: is a vector field that gives a *microscopic* measure of the rotation at any point in the fluid.
- A fluid flow having zero Vorticity is known as irrotational flow.

The Circulation Theorem

- The *circulation*, C , about a closed contour in a fluid is defined as the line integral evaluated along the contour of the component of the velocity vector that is locally tangent to the contour:

$$C \equiv \oint \mathbf{U} \cdot d\mathbf{l} = \oint |\mathbf{U}| \cos \alpha \, dl$$



$C > 0 \rightarrow$ Counterclockwise

$C < 0 \rightarrow$ Clockwise

Stokes' theorem

- In more general terms the relationship between **vorticity and circulation** is given simply by **Stokes' theorem** applied to the velocity vector:

$$\oint \mathbf{U} \cdot d\mathbf{l} = \iint_A (\nabla \times \mathbf{U}) \cdot \mathbf{n} dA$$

A area enclosed by the contour

n a unit vector normal to the area element **dA**

- Thus, Stokes' theorem states that the circulation about any closed loop is equal to the integral of the normal component of vorticity over the area enclosed by the contour

- Stokes' theorem is a general theorem in vector calculus that relates a *line integral of a vector field around a closed curve* to a *surface integral of the curl of that vector field over a surface bounded by the curve*.
- It is a fundamental theorem in mathematics and has applications in various fields, including fluid dynamics, electromagnetism, and differential geometry.

- Stokes' Theorem: It relates line integrals and surface integrals using the curl of the vector field. The theorem is expressed mathematically as
$$\oint(C) \mathbf{F} \cdot d\mathbf{r} = \iint(S) (\text{curl } \mathbf{F}) \cdot d\mathbf{S},$$
- where C is a closed curve, $\mathbf{F}$ is a vector field, $d\mathbf{r}$ represents a displacement vector along the curve, and $d\mathbf{S}$ represents an area vector on the surface.

Kelvin circulation theorem

- The Circulation Theorem, also known as Kelvin's Circulation Theorem, is a fundamental principle in fluid dynamics that relates *the circulation of a fluid flow around a closed curve* to *the rate of change of the fluid's vorticity within the enclosed region*.
- The theorem is named after the Scottish physicist William Thomson, 1st Baron Kelvin, who made significant contributions to the understanding of fluid dynamics.

- The Circulation Theorem is a crucial concept in the study of fluid motion and helps explain the behavior of vortices and the conservation of circulation in a fluid flow.
- The theorem can be stated as follows:
- The circulation around a closed curve in a steady, inviscid (non-viscous) flow is conserved, provided there are no sources or sinks of vorticity within the enclosed region.
- Mathematically, the theorem is expressed as:

$$C = \oint(C) \mathbf{v} \cdot d\mathbf{r}$$

Where:

C is the circulation around the closed curve C .

$\mathbf{v}$ is the velocity vector of the fluid.

$d\mathbf{r}$ represents an infinitesimal displacement vector along the curve.

Kelvin circulation theorem

- States that the circulation of a fluid parcel remains constant in the absence of external forces.

- In summary, the key difference lies in their focus and application. *Stokes' theorem is a general mathematical concept* used in various fields, including fluid dynamics, *whereas Kelvin's Circulation Theorem is specific to the conservation of circulation in fluid flows* and is particularly relevant in meteorology for understanding atmospheric and oceanic circulations.

The application of Circulation Theorem

- ☐ *The Circulation Theorem is applied for describing land and sea breeze.*
- ☐ *In meteorology and atmospheric science, Kelvin's Circulation Theorem is applied to explain the behavior of atmospheric vortices, such as cyclones and anticyclones.*
- ☐ *It helps describe the conservation of angular momentum and is used to analyze atmospheric circulations and the formation of weather systems.*

Various Forms of Vorticity

Vorticity and its Various Forms

Relative Vorticity: Vorticity associated with 3D shear in the wind field

$$\nabla \times \mathbf{V} \equiv \mathbf{i} \left(\frac{\partial \omega}{\partial y} - \frac{\partial v}{\partial p} \right) + \mathbf{j} \left(\frac{\partial u}{\partial p} - \frac{\partial \omega}{\partial x} \right) + \mathbf{k} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

Only the vertical component of vorticity (the **k** component) is of interest for large-scale (synoptic) meteorology

$$\xi = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Absolute Vorticity: The sum of relative and planetary vorticity

$$\eta \equiv \xi + f$$

Positive Vorticity: Associated with cyclonic (counterclockwise) circulations in the Northern Hemisphere

Negative Vorticity: Associated with anticyclonic (clockwise) circulations in the Northern Hemisphere

Vorticity and its Various Forms

- The difference between absolute and relative vorticity is *planetary vorticity*, which is just the local vertical component of the vorticity of the earth due to its rotation

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}, \quad \eta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} + f$$

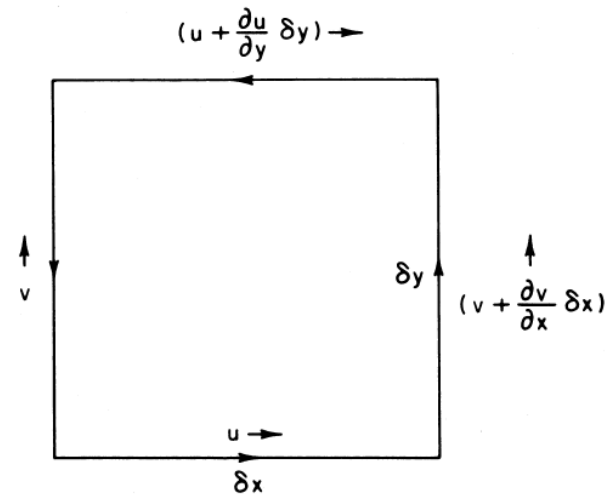
- The vertical component of vorticity is defined as the circulation about a closed contour in the horizontal plane divided by the area enclosed, in the limit where the area approaches zero:

$$\zeta \equiv \lim_{A \rightarrow 0} \left(\oint \mathbf{V} \cdot d\mathbf{l} \right) A^{-1}$$

Vorticity and its Various Forms

- considering the circulation about a rectangular element of area $\delta x \delta y$ in the (x, y) plane
- Evaluating $\mathbf{V} \cdot d\mathbf{l}$ for each side of the rectangle yields the circulation:

$$\begin{aligned} \delta C &= u \delta x + \left(v + \frac{\partial v}{\partial x} \delta x \right) \delta y - \left(u + \frac{\partial u}{\partial y} \delta y \right) \delta x - v \delta y \\ &= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \delta x \delta y \end{aligned}$$



- Dividing by the area $\delta A = \delta x \delta y$ gives

$$\frac{\delta C}{\delta A} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \equiv \zeta$$

Vorticity and its Various Forms

- In more general terms the relationship between vorticity and circulation is given simply by Stokes' theorem applied to the velocity vector:

$$\oint \mathbf{U} \cdot d\mathbf{l} = \iint_A (\nabla \times \mathbf{U}) \cdot \mathbf{n} dA$$

A area enclosed by the contour

$\mathbf{n}$ a unit normal to the area element dA

- Thus, Stokes' theorem states that the circulation about any closed loop is equal to the integral of the normal component of vorticity over the area enclosed by the contour

The Vorticity Equation

- We use the momentum equation to derive an equation for the time rate of change of vorticity.
- We differentiate the zonal component equation with respect to y and the meridional component equation with respect to x:

$$\frac{\partial}{\partial x} \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + fu = -\frac{1}{\rho} \frac{\partial p}{\partial y} \right) \quad \text{-----1}$$

$$\frac{\partial}{\partial y} \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} - fv = -\frac{1}{\rho} \frac{\partial p}{\partial x} \right) \quad \text{-----2}$$

- To compute the derivatives, all of the terms (except d/dt in each) require the use of the product rule for differentiation (quotient rule for last term of each expression). Doing so results in the following expanded derivatives of the momentum equations:

$$\begin{aligned}
 & \left[\underbrace{\left(\frac{\partial}{\partial x} \frac{\partial v}{\partial t} \right)}_{\text{green}} + \underbrace{\left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + u \frac{\partial^2 v}{\partial x^2} \right)}_{\text{cyan}} + \underbrace{\left(\frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + v \frac{\partial^2 v}{\partial x \partial y} \right)}_{\text{purple}} + \underbrace{\left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} + w \frac{\partial^2 v}{\partial x \partial z} \right)}_{\text{red}} + \right. \\
 & \quad \left. \underbrace{\left(f \frac{\partial u}{\partial x} + u \cancel{\frac{\partial f}{\partial x}} \right)}_{\text{yellow}} = - \frac{1}{\rho} \frac{\partial^2 p}{\partial x \partial y} + \frac{1}{\rho^2} \frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} \right] \text{-----3} \\
 & - \left[\underbrace{\left(\frac{\partial}{\partial y} \frac{\partial u}{\partial t} \right)}_{\text{green}} + \underbrace{\left(\frac{\partial u}{\partial y} \frac{\partial u}{\partial x} + u \frac{\partial^2 u}{\partial x \partial y} \right)}_{\text{cyan}} + \underbrace{\left(\frac{\partial v}{\partial y} \frac{\partial u}{\partial y} + v \frac{\partial^2 u}{\partial y^2} \right)}_{\text{purple}} + \underbrace{\left(\frac{\partial w}{\partial y} \frac{\partial u}{\partial z} + w \frac{\partial^2 u}{\partial y \partial z} \right)}_{\text{red}} - \right. \\
 & \quad \left. \underbrace{\left(f \frac{\partial v}{\partial y} + v \frac{\partial f}{\partial y} \right)}_{\text{yellow}} = - \frac{1}{\rho} \frac{\partial^2 p}{\partial x \partial y} + \frac{1}{\rho^2} \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y} \right] \text{-----4}
 \end{aligned}$$

- We can eliminate the term that involves a horizontal derivative of Coriolis, and subtract equation (equation3-equation4) (then combine the two equations into a single expression (boxes will now be used in grouping terms together):

$$\begin{aligned}
& \left(\frac{\partial}{\partial x} \frac{\partial v}{\partial t} \right) + \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + u \frac{\partial^2 v}{\partial x^2} \right) + \left(\frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + v \frac{\partial^2 v}{\partial x \partial y} \right) + \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} + w \frac{\partial^2 v}{\partial x \partial z} \right) + \left(f \frac{\partial u}{\partial x} \right) \\
& - \left(\frac{\partial}{\partial y} \frac{\partial u}{\partial t} \right) - \left(\frac{\partial u}{\partial y} \frac{\partial u}{\partial x} + u \frac{\partial^2 u}{\partial x \partial y} \right) - \left(\frac{\partial v}{\partial y} \frac{\partial u}{\partial y} + v \frac{\partial^2 u}{\partial y^2} \right) - \left(\frac{\partial w}{\partial y} \frac{\partial u}{\partial z} + w \frac{\partial^2 u}{\partial y \partial z} \right) + \left(f \frac{\partial v}{\partial y} + v \frac{\partial f}{\partial y} \right) \quad \text{-----5} \\
& = -\frac{1}{\rho} \frac{\partial^2 p}{\partial x \partial y} + \frac{1}{\rho^2} \frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} + \frac{1}{\rho} \frac{\partial^2 p}{\partial x \partial y} - \frac{1}{\rho^2} \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y}
\end{aligned}$$

- We can eliminate the second derivative pressure terms (red boxes above). We will also combine the Eulerian u/v tendency terms (and switch the order of differentiation, green boxes) and combine the u, v, and w terms (purple) to get:

$$\begin{aligned}
& \left(\frac{\partial}{\partial t} \frac{\partial v}{\partial x} - \frac{\partial}{\partial t} \frac{\partial u}{\partial y} \right) + u \left(\frac{\partial^2 v}{\partial x^2} - \frac{\partial^2 u}{\partial x \partial y} \right) + v \left(\frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 u}{\partial y^2} \right) + w \left(\frac{\partial^2 v}{\partial x \partial z} - \frac{\partial^2 u}{\partial y \partial z} \right) + \\
& \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} \right) + \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \left(f \frac{\partial u}{\partial x} + f \frac{\partial v}{\partial y} \right) + v \frac{\partial f}{\partial y} \quad \text{-----6} \\
& = \frac{1}{\rho^2} \left(\frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} - \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y} \right)
\end{aligned}$$

- Next, pull out the d/dt in the first term, d/dx in the second term, d/dy in the third term, and d/dz in the fourth term, and bring out the Coriolis parameter in the term in the second row (blue):

$$\begin{aligned}
 & \frac{\partial}{\partial t} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + u \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + v \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + w \frac{\partial}{\partial z} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + \\
 & \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} \right) + \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + f \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + v \frac{\partial f}{\partial y} \quad \text{-----7} \\
 & = \frac{1}{\rho^2} \left(\frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} - \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y} \right)
 \end{aligned}$$

- For the first four terms, we will replace (dv/dx-du/dy) with zeta/ vorticity, and rewrite the first term of the second row in terms of vorticity times divergence:

$$\left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} \right) = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \quad \text{Refer to equation 5.}$$

To get to:

$$\begin{aligned}
 & \frac{\partial \zeta}{\partial t} + u \frac{\partial \zeta}{\partial x} + v \frac{\partial \zeta}{\partial y} + w \frac{\partial \zeta}{\partial z} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + f \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + v \frac{\partial f}{\partial y} \quad \text{-----8} \\
 & = \frac{1}{\rho^2} \left(\frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} - \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y} \right)
 \end{aligned}$$

- We can define the Lagrangian derivative of Coriolis as: $\frac{df}{dt} = \cancel{\frac{\partial f}{\partial t}} + u \cancel{\frac{\partial f}{\partial x}} + v \frac{\partial f}{\partial y} + w \cancel{\frac{\partial f}{\partial z}}$

- But we can take advantage of the fact that the only **non-zero** term is the term that contains the **meridional** derivative and rewrite this as:

$$\frac{df}{dt} = v \frac{\partial f}{\partial y}$$

- We combine the Eulerian and advective derivatives for relative vorticity (green and purple) (equation 8) into the total derivative, and replace the $v(df/dy)$ term, we can condense the previous expression as (equation 8):

$$\frac{d\xi}{dt} + \xi \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + f \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \frac{df}{dt} = \frac{1}{\rho^2} \left(\frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} - \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y} \right) \quad \text{-----9}$$

- Combining the relative vorticity and Coriolis into a single total derivative and combining the terms that contain a multiplication of the divergence:

$$\frac{d(\xi + f)}{dt} + (\xi + f) \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) = \frac{1}{\rho^2} \left(\frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} - \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y} \right) \quad \text{-----10}$$

- Finally, we move everything to the RHS to come up with the vorticity equation!

$$\frac{d(\zeta + f)}{dt} = \overset{1}{-\left(\zeta + f\right)\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right)} - \overset{2}{\left(\frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z}\right)} + \overset{3}{\frac{1}{\rho^2} \left(\frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} - \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y}\right)} \quad \text{-----11}$$

This (equation 11) is the vorticity equation in height coordinates.

- This equation states that the Lagrangian tendency (time rate of change following the flow) of the absolute vorticity consists of:

1. divergence term

2. tilting term

3. solenoid term

Example

If wind rotates as a solid body around the centre of a low-pressure system and the tangential velocity is 10 m s^{-1} at a radius of 200 km, find the relative vorticity.

Example

A 10km deep layer of air at 45 degree latitude with speed of 10m/s across a distance of 300km. What is the potential vorticity ?

....END

Thanks !

1. Which of the following statements about vorticity is true?
 - A. Vorticity is always positive.
 - B. Vorticity is always negative.
 - C. Vorticity can be either positive or negative.
 - D. Vorticity is not related to the rotation of fluid particles.
2. What is the relationship between circulation and vorticity?
 - A. Circulation is the gradient of vorticity.
 - B. Circulation is the divergence of vorticity.
 - C. Circulation is the curl of vorticity.
 - D. Circulation is the Laplacian of vorticity
3. What is the mathematical expression for circulation?

A. $\oint \mathbf{V} \cdot d\mathbf{r}$	C. $\nabla \times (\mathbf{V} \times \nabla \times \mathbf{V})$
B. $\nabla \times \mathbf{V}$	D. $\nabla \times (\mathbf{V} \times \mathbf{B})$

1. What is the Kelvin circulation theorem?
 - A. States that the circulation of a fluid parcel remains constant in the absence of external forces.
 - B. States that the circulation of a fluid parcel decreases in the presence of external forces.
 - C. States that the circulation of a fluid parcel increases in the presence of external forces.
 - D. States that the circulation of a fluid parcel is dependent on its mass.
2. What is the mathematical expression for absolute vorticity?

A) $\partial v / \partial x - \partial u / \partial y + f$	C) $\partial v / \partial y - \partial u / \partial x + f$
B) $\partial u / \partial x - \partial v / \partial y + f$	D) $\partial u / \partial y - \partial v / \partial x + f$
3. What is the mathematical expression for relative vorticity?

A) $\partial u / \partial y - \partial v / \partial x$	C) $\partial v / \partial y - \partial u / \partial x$
B) $\partial u / \partial x - \partial v / \partial y$	D) $\partial v / \partial x - \partial u / \partial y$

1. What is the purpose of using simplified equations and models to describe atmospheric behavior?
 - A) To make predictions and understand the behavior of the atmosphere
 - B) To make the equations easier to solve analytically
 - C) To simplify the physical laws governing atmospheric phenomena
 - D) To reduce the complexity and scale of atmospheric phenomena
2. What is the planetary vorticity?
 - A. The vorticity due to the rotation of the Earth.
 - B. The vorticity due to the Earth's gravitational pull.
 - C. The vorticity due to atmospheric turbulence.
 - D. The vorticity due to the Coriolis Effect.

1. What is the relationship between circulation and vorticity?
 - A. Circulation is the gradient of vorticity.
 - B. Circulation is the divergence of vorticity.
 - C. Circulation is the curl of vorticity.
 - D. Circulation is the Laplacian of vorticity.

2. Which theorem describes the conservation of absolute vorticity in a non-divergent flow?
 - A) Kelvin's circulation theorem
 - B) Helmholtz's vorticity theorem
 - C) Bernoulli's equation
 - D) The continuity equation

1. What is the definition of vorticity in atmospheric dynamics?
 - A) The total mass of air in a given region
 - B) The rate at which air moves vertically in a given region
 - C) The tendency of a fluid to rotate about an axis
 - D) The measure of the density of air in a given region
2. Which of the following is not a key application of potential vorticity in atmospheric dynamics?
 - A) Synoptic meteorology
 - B) Numerical weather prediction
 - C) Climate modeling
 - D) Surface weather observations
3. What is the approximate value of planetary vorticity at the equator?
 - A) $5 \times 10^{-5} \text{ s}^{-1}$
 - B) $2 \times 10^{-5} \text{ s}^{-1}$
 - C) 0
 - D) 10^{-4} s^{-1}